Blob Detection

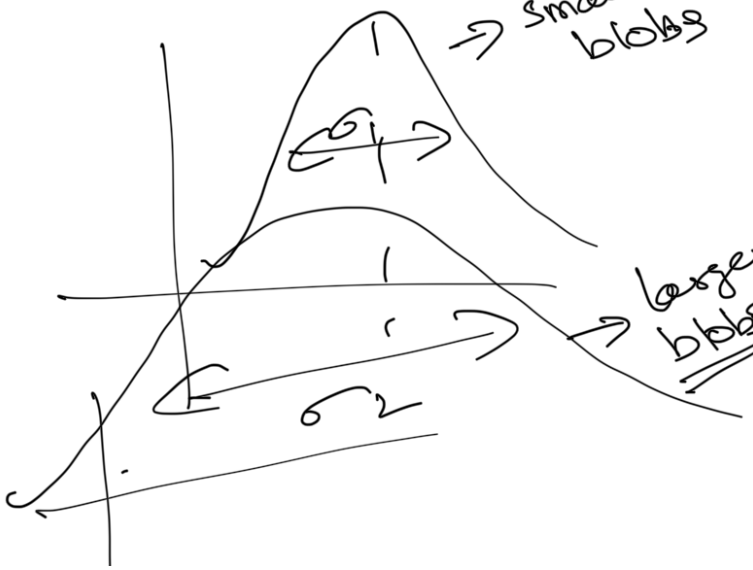
CV ⇒ models blobs

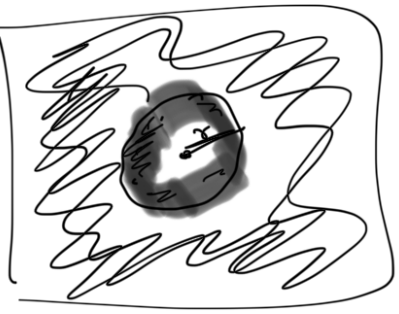
Assumption 1: as circular region with constant or similar intensities.

(light on dark or dark on light)



Assumption 2: Intensity changes gradually in form of a 2-D Gaussian





radius depends on σ

2D Gaussian

$$\Rightarrow f(x,y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right)$$

$\sigma \Rightarrow$ standard deviation

Separable function

$$f(x,y) = f(x)\,f(y) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-x^2}{2\sigma^2}\right)\right)\left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-y^2}{2\sigma^2}\right)\right)$$

Kernel $\Rightarrow$ Gaussian Kernel

$$\begin{matrix} f(-1,1) & f(0,1) & f(1,1) \\ f(-1,0) & f(0,0) & f(1,0) \\ f(-1,-1) & f(0,-1) & f(1,-1) \end{matrix}$$

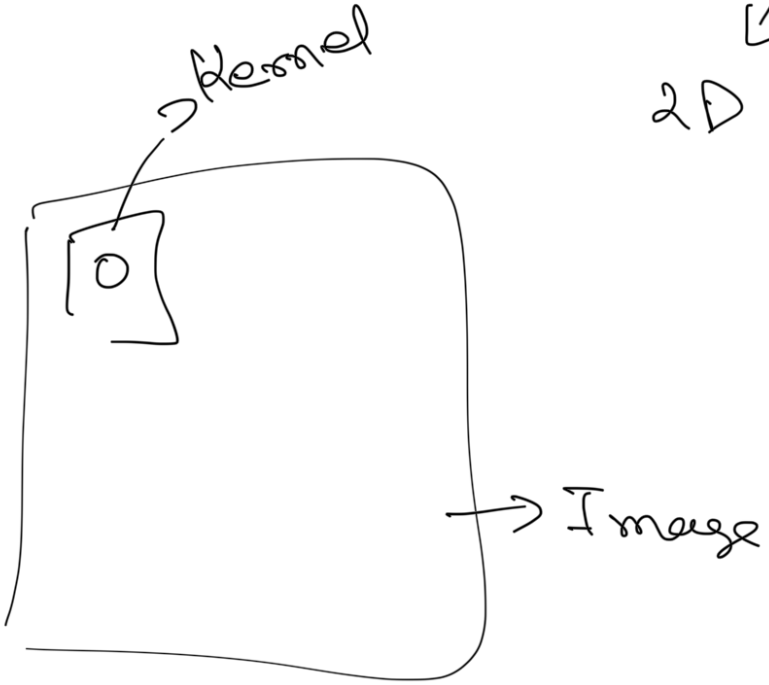
3x3

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Smoothening filter

$\rightarrow$ Gaussian Kernel

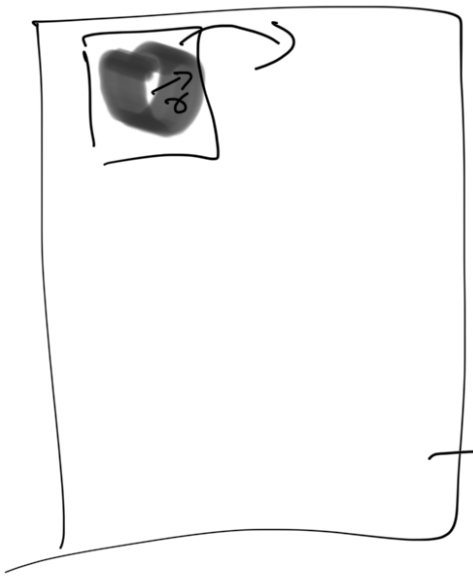
$\rightarrow$ 2D Convolution or Spatial filter



=> Slide the Gaussian Kernel

=> Wherever the center pixel value is maximum (this point becomes center of blob).

=> Take various values of Sigma ⇒ find various blobs with varying radii.



Scale Space:

Kernel 1 = $f(x, y, \sigma)$

Kernel 2 = $f(x, y, k\sigma)$

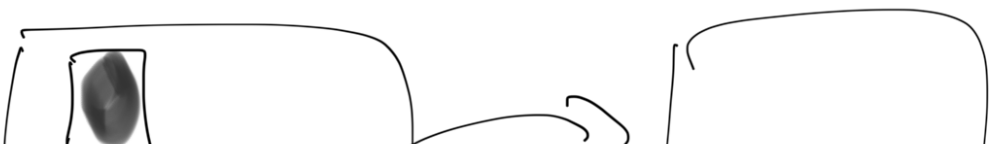
Kernel 3 = $f(x, y, 2k\sigma)$

Kernel 4 = $f(x, y, 3k\sigma)$

$\vdots$

$k = \sqrt{2}$

Experimentally determined





smoothed image

where ~~centers~~ blobcenters will have maximum intensity

Laplacian filter (edge detection detects the center of the blob).

Input Image $I(x,y)$

$$O(x,y) = L(x,y) * \left(G(x,y,\sigma) * I(x,y) \right)$$

$$= LoG(x,y,\sigma) * I(x,y)$$

Laplacian of Gaussian.

In the output image, blob center are given as white dots.

radius?

$$G(x,y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x^2+y^2)}{2\sigma^2}\right)$$

$$= \frac{1}{2\pi\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right)$$

$$G(0,0) = \frac{1}{2\pi\sigma^2}$$



radius is that value where Gauss ian decrease by e^{-1} to the value at center

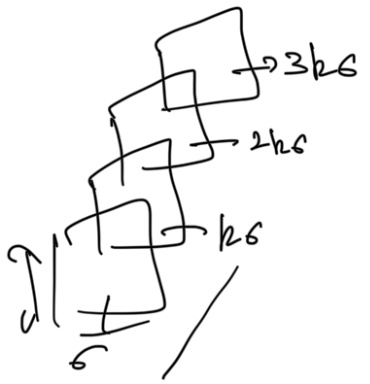
$$\frac{r^2}{2\sigma^2} = 1$$

$$r = \sqrt{2}\sigma$$

Steps for Blob Detection using

LoG

Step 1: Apply LoG at various scales ($\sigma, k\sigma, 2k\sigma, \ldots$)



↓

The Generates a Scale Space.

↓

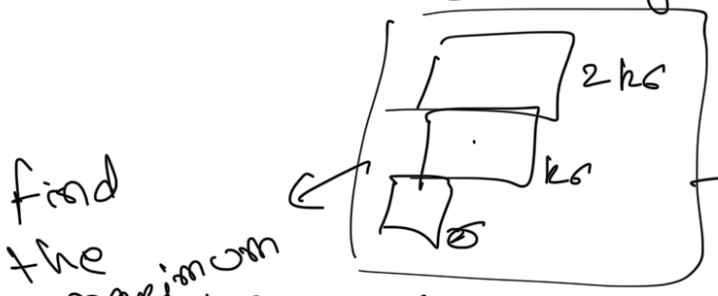
2D Image is now converted to collection of Image

Step 2: Find Maximum Points in a neighboring set => 3x3x3

Kernel => 3x3

-1

$\begin{bmatrix} \cdot \end{bmatrix}$



find the maximum value in these three images.

Center Pixel should be larger than neighboring 26 pixels

Step 3: If center pixel is larger than neighbouring 26 pixels then the center pixel becomes blob center

↳ The radius of the blob is $(\sqrt{2}(k\sigma))$

⇒ One Drawback ⇒ Finding so many LoG is Computationally intensive.

↓

Another method of finding LoG ⇒

$$G(x,y,\sigma) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x^2+y^2)}{2\sigma^2}\right)$$

$$\ln G(x,y,\sigma) = -\left(\frac{x^2+y^2}{2\sigma^2}\right) - \ln 2\pi - \ln\sigma^2$$

$$\frac{\partial \ln G(x,y,\sigma)}{\partial(\sigma^2)} = \frac{x^2+y^2}{2(\sigma^2)^2} - \frac{1}{\sigma^2}$$

$$\boxed{\frac{\partial \ln G(x,y,\sigma)}{\partial(\sigma^2)} = \frac{(x^2+y^2) - 2\sigma^2}{2\sigma^4}} \quad (1)$$

$$\frac{1}{G}\frac{\partial G}{\partial(\sigma^2)} = \frac{(x^2+y^2) - 2\sigma^2}{2\sigma^4}$$

$$\frac{\partial G}{\partial \sigma^2} = \left(\frac{(x^2+y^2) - 2\sigma^2}{2\sigma^4}\right) G \quad (1)$$

LoG

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$\text{Laplacian} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

$$LoG\,(I(x,y)) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) G * (I(x,y))$$

$$= \left(\frac{\partial^2 G}{\partial x^2} + \frac{\partial^2 G}{\partial y^2}\right) * I(x,y)$$

$$G = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x^2+y^2)}{2\sigma^2}\right)$$

$$\frac{\partial G}{\partial x} = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x^2+y^2)}{2\sigma^2}\right) \times -\frac{1}{2\sigma^2} \times 2x$$

$$\frac{\partial G}{\partial x} = -\frac{x}{\sigma^2} G$$

$$\frac{\partial^2 G}{\partial x^2} = -\frac{G}{\sigma^2} - \frac{x}{\sigma^2} \times \frac{\partial G}{\partial x}$$

$$= -\frac{G}{\sigma^2} - \frac{x}{\sigma^2} \times \frac{-xG}{\sigma^2}$$

$$= \frac{G}{\sigma^2}\left(\frac{x^2}{\sigma^2} - 1\right)$$

$$\frac{\partial^2 G}{\partial x^2} = \frac{G}{\sigma^2}\left(\frac{x^2 - \sigma^2}{\sigma^2}\right)$$

$$\frac{\partial^2 G}{\partial y^2} = \frac{G}{\sigma^2}\left(\frac{y^2 - \sigma^2}{\sigma^2}\right)$$

$$LoG = \frac{\partial^2 G}{\partial x^2} + \frac{\partial^2 G}{\partial y^2} = \frac{G}{\sigma^4}\left(x^2 + y^2 - 2\sigma^2\right) \quad (IV)$$

$$LoG = \frac{1}{2}\frac{\partial G}{\partial(\sigma^2)}$$

$$LoG \approx \frac{1}{2}\,\frac{G(x, y, k\sigma) - G(x, y, \sigma)}{k^2\sigma^2 - \sigma^2}$$

$$LoG = \frac{1}{2\sigma^2} \frac{G(x, y, k\sigma) - G(x, y, \sigma)}{k^2 - 1}$$

Magnitude is inversely related to σ. Difficult to detect large blobs

This technique of finding LoG is know Difference of Gaussian Method.

Solution ↓ Use a Normalized LoG:

Normalized LoG

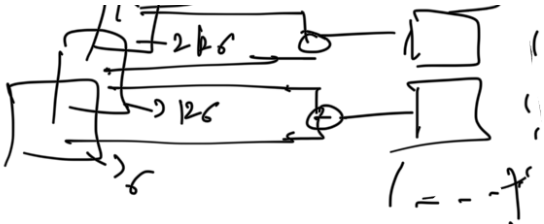
$$(\sigma^2 LoG) \approx \frac{G(x, y, k\sigma) - G(x, y, \sigma)}{k^2 - 1}$$

Steps for Blob Detection using DoG

Step1: Find Gaussian smoothed image at various scales $(\sigma, k\sigma, 2k\sigma \ldots)$

Step2: Find Difference of Gaussians





New Neighbourhood DoG

Step3: Find Maximum in the neighbourhood obtained in step2.

Step4: The maximum obtained becomes blob center.

Radius is taken as average of neighboring scale $\Rightarrow x = \sqrt{2}\left(\frac{\sigma + k\sigma}{2}\right)$